



## Shiv Heart AI

Machine Learning Model

### About

This application predicts the likelihood of Heart Disease using Machine Learning.

### Model Used

✓ K-Nearest Neighbors

### Developer

Shiv Shankar Tiwari

### Technology

- Python
- Scikit-Learn
- Streamlit
- Pandas

This application was developed by Shiv Shankar Tiwari as a Machine Learning project to demonstrate predictive analytics using Python, Scikit-learn, and Streamlit. It is intended for learning, research, and portfolio demonstration purposes only and should not be used as a substitute for professional medical diagnosis or treatment.

# Shiv ❤️ Heart Disease Prediction AI

Machine Learning Based Cardiovascular Risk Assessment System



## Patient Information

Please enter the patient's clinical information.



### Personal Details

Age



Gender

☒ M ☐ F

Chest Pain Type

ATA

Resting Blood Pressure (mmHg)

120

Cholesterol (mg/dL)

200



### Clinical Information

Fasting Blood Sugar (>120 mg/dL)

0

Resting ECG

Normal

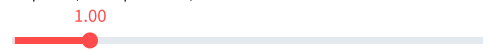
Maximum Heart Rate



Exercise Induced Angina

☒ N ☐ Y

Oldpeak (ST Depression)



ST Slope

Up



### Patient Summary

Age

40 Years

Cholesterol

200 mg/dL

Gender

Male

Blood Pressure

120 mmHg

Maximum HR

150

Fasting Sugar

Normal



Predict Heart Disease Risk



### Prediction Result

✓

LOW RISK OF HEART DISEASE

Healthy Probability  
100.00%

AI Confidence

Personalized Health Recommendations

Great News!

Your prediction indicates a lower risk of heart disease.

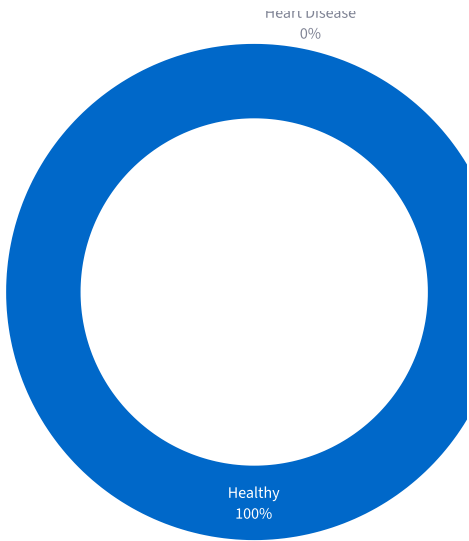
Maintain these healthy habits:

- Exercise regularly.
- Eat a balanced diet.
- Stay hydrated.
- Sleep 7–8 hours.
- Avoid smoking.
- Continue annual health checkups.
- Maintain a healthy BMI.

Keep taking care of your heart ❤️

AI Analysis Dashboard

Prediction Probability



Risk Level

Risk Percentage  
0.0%

Low Risk

Heart Health Score

Health Score  
100/100



# Patient Report

| Feature             | Value  |
|---------------------|--------|
| Age                 | 40     |
| Gender              | M      |
| Chest Pain          | ATA    |
| Blood Pressure      | 120    |
| Cholesterol         | 200    |
| Fasting Blood Sugar | 0      |
| Rest ECG            | Normal |
| Maximum HR          | 150    |
| Exercise Angina     | N      |
| OldPeak             | 1.0    |

 Download Prediction Report



# Heart Health Tips

 **Healthy Diet**

- Eat fruits
- Green vegetables
- Whole grains
- Reduce sugar
- Reduce salt

 **Physical Activity**

- Walk daily
- Exercise 30 min
- Yoga
- Cycling
- Swimming

 **Medical Care**

- Annual Check-up
- Blood Pressure
- Cholesterol Test
- Blood Sugar Test



# Machine Learning Model

|           |                  |            |
|-----------|------------------|------------|
| Algorithm | Training Samples | Prediction |
| KNN       | 918              | Binary     |



# Heart Disease Prediction AI

Developed by Shiv Shankar Tiwari

Python • Streamlit • Scikit-Learn • Machine Learning

 This application is intended for getting my personal work expertise.

